

# A Mean Field View of the Landscape of Two-Layer Neural Networks

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## Distributional Dynamics (DD)

The fundamental PDE governing the infinite-width limit of two-layer neural networks:

$$\partial_t \rho_t = 2\xi(t) \nabla_\theta \cdot (\rho_t \nabla_\theta \Psi(\theta; \rho_t)), \quad (1)$$

where the potential  $\Psi$  is defined as

$$\Psi(\theta; \rho) = V(\theta) + \int U(\theta, \theta') \rho(d\theta'), \quad (2)$$

with  $V(\theta) = -\mathbb{E}[y \sigma_*(x; \theta)]$  and  $U(\theta_1, \theta_2) = \mathbb{E}[\sigma_*(x; \theta_1) \sigma_*(x; \theta_2)]$ .

## Main Convergence Result

**Theorem 1** (**Theorem 3** in paper: SGD  $\rightarrow$  PDE). *Assume conditions A1–A3 (bounded Lipschitz step size, bounded activation with sub-Gaussian gradient, bounded Lipschitz gradients of  $V$  and  $U$ ). For  $\rho_0 \in \mathcal{P}(\mathbb{R}^D)$ , consider SGD with initialization  $(\theta_i^0)_{i \leq N} \stackrel{iid}{\sim} \rho_0$  and step size  $s_k = \varepsilon \xi(k\varepsilon)$ . For  $t \geq 0$ , let  $\rho_t$  be the solution of the DD PDE. Then for any fixed  $t$ , the empirical measure  $\hat{\rho}_t^{(N)} \Rightarrow \rho_t$  almost surely along any sequence  $(N, \varepsilon = \varepsilon_N)$  such that  $N/\log(1/\varepsilon_N) \rightarrow \infty$  and  $\varepsilon_N \rightarrow 0$ .*

*Moreover, there exists a constant  $C$  (depending only on the parameters in A1–A3) such that for any  $f : \mathbb{R}^D \rightarrow \mathbb{R}$  with  $\|f\|_\infty, \|f\|_{Lip} \leq 1$  and  $\varepsilon \leq 1$ ,*

$$\sup_{k \in [0, T/\varepsilon] \cap \mathbb{N}} \left| \frac{1}{N} \sum_{i=1}^N f(\theta_i^k) - \int f(\theta) \rho_{k\varepsilon}(d\theta) \right| \leq C e^{CT} \cdot err_{N,D}(z), \quad (3)$$

$$\sup_{k \in [0, T/\varepsilon] \cap \mathbb{N}} \left| R_N(\theta^k) - R(\rho_{k\varepsilon}) \right| \leq C e^{CT} \cdot err_{N,D}(z), \quad (4)$$

*with probability at least  $1 - e^{-z^2}$ , where the error term is*

$$err_{N,D}(z) = \sqrt{\frac{1}{N} \vee \varepsilon \cdot (\sqrt{D} + \log(N/\varepsilon) + z)}. \quad (5)$$

**Significance.** This theorem establishes that SGD dynamics does not become more complex as the network width  $N$  increases. The approximation is valid when  $N \gg D$  and  $\varepsilon \ll 1/D$ . The error grows exponentially with the time horizon  $T$ , but is benign in  $N$  and  $D$ .

# 1 Dependency Graph

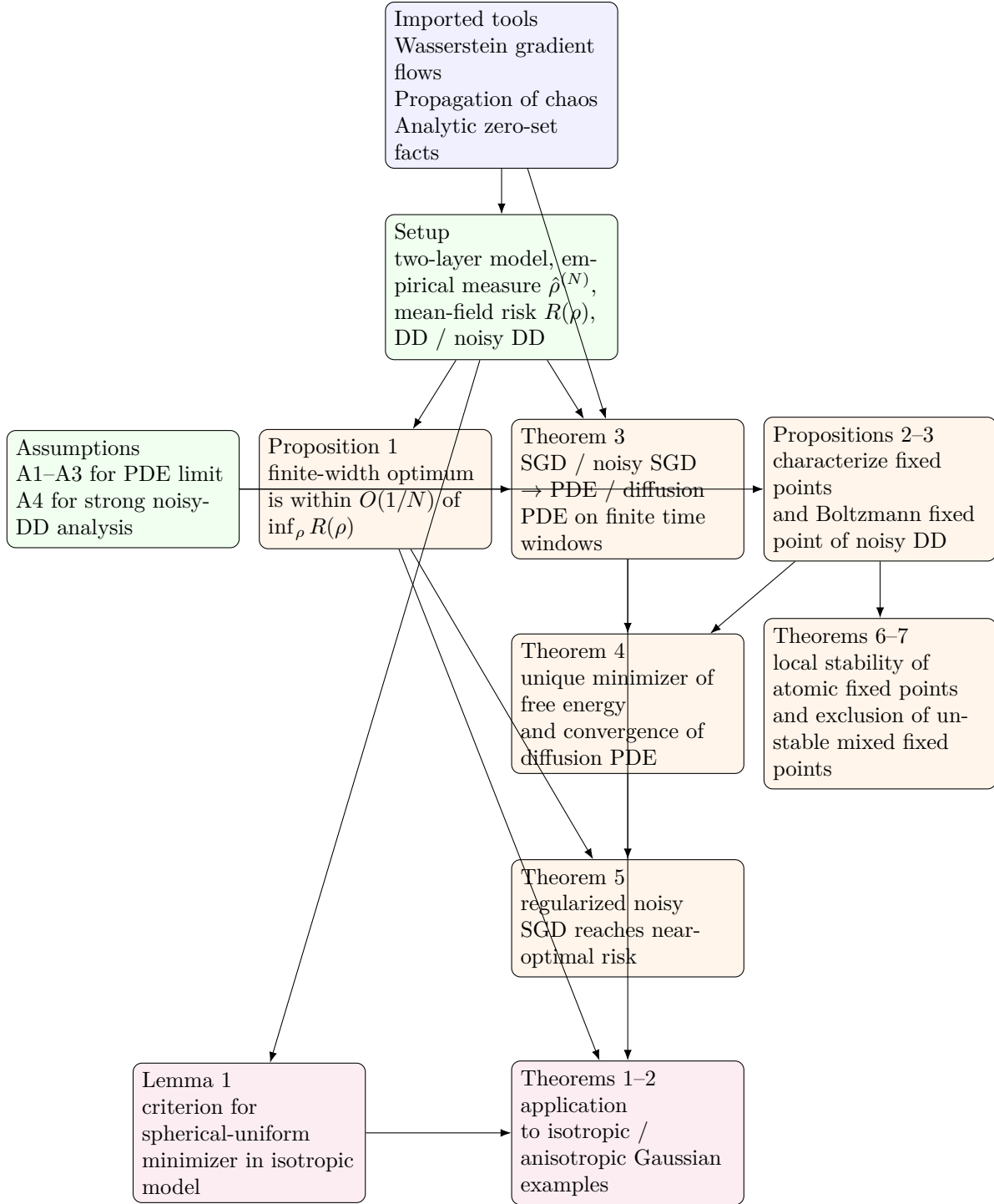


Figure 1: Main proof spine of [1]. The central transition is from finite-width SGD to distributional dynamics, after which the analysis moves to fixed points, free energy, and example-specific PDE reductions. A few logically true but visually redundant edges are omitted so the main proof spine remains readable.

**How to read the graph.** The paper’s most important move is Theorem 3: once SGD is approximated by DD, the optimization problem is replaced by the study of a measure-valued flow. The noisy branch (Proposition 3  $\rightarrow$  Theorem 4  $\rightarrow$  Theorem 5) gives the cleanest generic convergence result. The noiseless branch (Propositions 2, Theorems 6–7) gives a weaker but structurally informative picture of stable and unstable fixed points.

## References

- [1] Song Mei, Andrea Montanari, and Phan-Minh Nguyen. A mean field view of the landscape of two-layer neural networks. *Proceedings of the National Academy of Sciences*, 115(33):E7665–E7671, 2018.